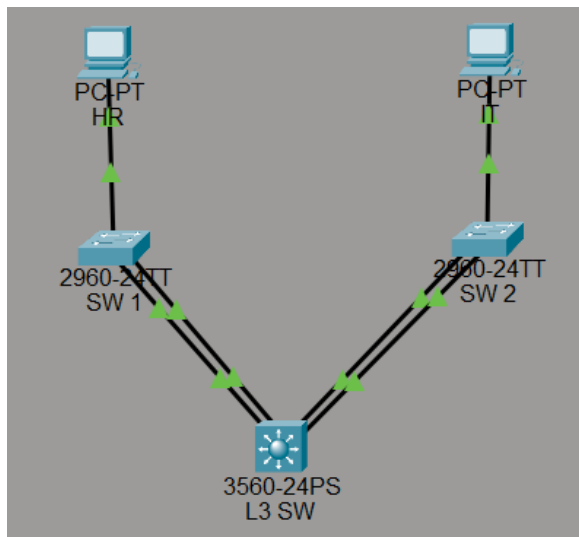


MINI PROJECT 7

Layer-3 Switch Inter-VLAN Routing + LACP EtherChannel + Trunking

DIAGRAM:



DESCRIPTION:

This project explains how a Layer-3 switch performs Inter-VLAN routing to allow communication between different broadcast domains. It also shows how LACP EtherChannel and trunking are used to provide higher bandwidth, redundancy, and efficient VLAN traffic between switches.

STEP 1: Create VLANs on SW 1,SW2:

Verify existing VLANs from Previous projects

STEP 2: Assign Access Ports to VLANs on SW1 & SW2:

SW1:

```
interface fastEthernet0/1
switchport mode access
switchport access vlan 10
exit
```

SW 2:

```
interface fastEthernet0/1
switchport mode access
switchport access vlan 20
exit
```

STEP 3: Configure LACP EtherChannel

Sw1 & L3 switch,

```
interface range fastEthernet0/21 - 22
channel-group 1 mode active
exit
```

sw2 & L3 Switch,

```
interface range fastEthernet0/23 - 24
channel-group 2 mode active
exit
```

passive on both sides will not create etherchannel

STEP 4 : Configure Port-Channel as Trunk:

Sw1 & L3 switch:

```
interface port-channel 1
switchport mode trunk
switchport trunk allowed vlan 10,20
exit
```

sw2 & L3switch:

```
interface port-channel 2
switchport mode trunk
switchport trunk allowed vlan 10,20
exit
```

STEP 5: Configure IP Addressing on PCs:

PC	VLAN	IP Address	Gateway
PC-HR	10	192.168.10.10/24	192.168.10.1
PC-IT	20	192.168.20.10/24	192.168.20.1

STEP 6: Verification & STP Observation:

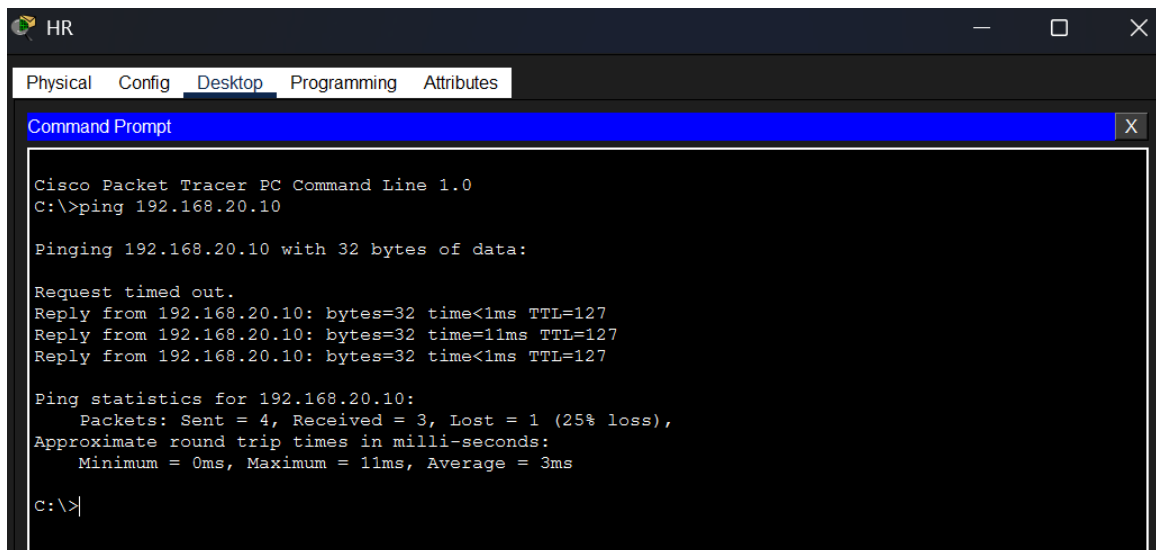
show etherchannel summary on L3 Switch,

```
Number of channel-groups in use: 2
Number of aggregators:          2

Group  Port-channel  Protocol    Ports
-----+-----+-----+-----
1      Po1 (SU)          LACP        Fa0/21 (P) Fa0/22 (P)
2      Po2 (SU)          LACP        Fa0/23 (P) Fa0/24 (P)
Switch#
```

show spanning-tree

STEP 7: Ping Test:



```
HR
Physical Config Desktop Programming Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.20.10

Pinging 192.168.20.10 with 32 bytes of data:

Request timed out.
Reply from 192.168.20.10: bytes=32 time<1ms TTL=127
Reply from 192.168.20.10: bytes=32 time=11ms TTL=127
Reply from 192.168.20.10: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.20.10:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 11ms, Average = 3ms

C:\>
```

We have Layer 3 switch, so we can ping from HR—> IT PC .